Duckophila: a whole-brain *Drosophila* connectome model in closed loop with a bipedal robot

Technical report and preprint draft

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Abstract

We connect the complete connectome-derived leaky integrate-and-fire model of the adult *Drosophila melanogaster* brain (FlyWire, 138,639 neurons, 54.5 million synapses; Shiu et al. 2024) to a small bipedal robot, the open-source microduck of Pollen Robotics (Hugging Face), running its production control software against a MuJoCo simulation of its body. The robot's time-of-flight depth sensor is translated into input to the fly's looming-sensitive visual projection neurons (LPLC2, LC4); its camera, through dense optic flow, into input to the fly's elementary motion detectors (T4/T5), each placed at its own visual column using FlyWire's column map. Motor commands are read out from identified descending neurons. Without any tuning of the connectome, the model (i) fires its giant fibre (DNp01) to objects approaching the robot's face (20 of 20 throws in closed loop), and (ii) produces a syndirectional optomotor response in DNp15/DNHS1 that turns the robot's head—and, through a coupling rule of ours, its walking body—with a rotating striped drum, at a closed-loop gain of 0.23 [0.21, 0.27], and (iii) turns the walking robot toward a small target through LC10a, the anterior optic tubercle and DNa02, keeping up with a target led 90° each way in 4 of 8 runs (heading gain 1.00). All three require the wiring: a degree-preserving rewired connectome with the same neurons and weights produces none (0 of 20; gain 0.00; 0 of 8). Further controls show that DNp15 is the strongest rotation-following neuron of 383 paired descending neuron types, while the escape response is only partly specific to the looming detectors and only weakly lateralised, and that the target pathway steers only for targets below the eye's equator, and more weakly to the right. The model's mushroom body, in contrast, has no input-specific code: any input that reaches it recruits the same two-thirds of its Kenyon cells, through a runaway that starts in the antennal lobe. With cue codes, a dopamine plasticity rule and a valence readout supplied by us, the robot learns to approach whichever of two targets it was petted for (5 of 6 runs; the other, 0 of 6). With a motion-based encoder for a real room the same pathway turns the robot after a person's feet (median heading error 19° against 40° rewired), though only with a readout, a correction and control rules of ours, listed. The brain can also drive the robot from a laptop over WebRTC, with every camera frame dated on the robot's clock. To run the model in closed loop we reimplemented it as a stepped simulator that reproduces the reference Brian2 implementation spike for spike and runs faster than real time on one CPU core, using an exact closed-form bound that lets neurons unable to fire before their next input remain unsimulated. We report what the connectome contributes, what our hand-written sensory and motor interfaces contribute, and several failure modes found on the way: a dropped-input rule of the reference model, scrambled retinotopy destroying LPLC2 selectivity, the robot's own gait dominating its optic flow, and a field of view too narrow for the canonical steering neuron DNa02. Code, experiments and a recorded demonstration are open source.

1 Introduction

Complete synaptic wiring diagrams of an adult insect brain now exist. FlyWire reconstructed the whole brain of a female *Drosophila* [1], annotated every neuron with a cell type [2], and mapped the visual system to its retinotopic columns [3]. Shiu et al. [4] showed that a leaky integrate-and-fire (LIF) network built directly from this connectome—every synapse at one shared weight, its sign taken from the predicted neurotransmitter—predicts sensorimotor pathways that were then confirmed experimentally, for example from sugar-sensing gustatory neurons to the proboscis motor neuron MN9.

That model has so far been driven by prescribed stimulation and read out offline. Here we ask a different question: what does it do when its sensory neurons are driven by a robot's sensors, its descending neurons move that robot, and the consequences of those movements come back through the sensors—in closed loop,

Table 1: What is the connectome's and what is ours. The brain model is not trained, tuned or edited.

From the connectome (unchanged)	Hand-written interfaces (ours)
Every neuron, synapse and sign; the LIF dynamics of [4]	ToF → LPLC2/LC4 Poisson rates (inverse time-to-contact)
Which neurons respond to looming, which to rotation	Camera → optic flow → T4/T5 rates by preferred direction
That the giant fibre fires (and more reliably for LPLC2/LC4 input)	The shape of the startle (sound, gaze, crouch, step back)
The sign and size of the DNp15/DNa02 left–right difference	Gains, smoothing, clamps; the body following the head
Where each T4/T5 looks (FlyWire columns); front and up, from the wiring	Approach-speed gate; efference copies of the robot's own motion
That LC10a → AOTU019/025 → DNa02 turns toward a small target, only below the equator	Target detector; bearing → LC10a rates and their dose; walking while a target is seen
Which DN pairs carry a target's bearing; that two feet steer and whole legs do not	For a person: motion pooled onto LC10a receptive fields, its lowest 8°; a four-DN readout and its lean; heading memory, suppression, pursuit gate
Kenyon cells → MBONs → downstream; which DANs reach which MBON	Learning: cue codes, plasticity rule, valence by compartment, antennal lobe silenced, approach gate

in real time? The robot is the microduck, a 15-servo bipedal robot from Pollen Robotics with an 8×8 time-of-flight (ToF) depth sensor and a head camera, controlled by reinforcement-learned locomotion policies. We do not replace the robot's locomotion: the fly brain sits above it, as a behaviour source, the way descending neurons sit above the fly's ventral nerve cord.

Our contribution is an engineering study with three parts: (1) a stepped, exact and faster-than-real-time reimplementation of the Shiu et al. model; (2) sensory interfaces that respect the fly's own anatomy—in particular a retinotopic placement of every T4/T5 neuron whose orientation is derived from the connectome itself; and (3) closed-loop experiments in a simulated twin of the robot that runs the robot's production daemons. Throughout, we separate what the connectome does from what our interfaces do, because that boundary is what any claim about "a fly brain controlling a robot" depends on (Table 1).

2 Methods

2.1 Brain model and a stepped reimplementation

We use the FlyWire 783 connectome as distributed with [4] (138,639 neurons, 15,091,983 connected pairs, 54,492,922 synapses) and the model's equations and constants unchanged: membrane time constant 20 ms, synaptic time constant 5 ms, rest and reset −52 mV, threshold −45 mV, refractory period 2.2 ms, synaptic delay 1.8 ms, 0.275 mV per synapse, time step 0.1 ms, and Poisson stimulation as a voltage kick of 250 × 0.275 mV.

The reference implementation in Brian2 [5] rebuilds the network for every run, which is appropriate for one-second optogenetic trials but not for a sensorimotor loop that needs tens of milliseconds of brain time per sensor frame (a 20 ms step cost 0.2 s to 0.4 s of wall time; a long run, 2.6× real time). Our stepped simulator (`flysim.py`, numba) holds the state between calls and follows Brian2's schedule within each step: state update, threshold, synaptic propagation and Poisson input, reset.

A semantic detail of the reference. In the reference model v and g are declared `(unless refractory)`. Brian2 implements this as a *conditional write*: any write to these variables, including synaptic input, is discarded while the neuron is refractory. Input arriving during refractoriness is therefore lost, not accumulated. An implementation that banks it is plausible-looking but wrong: every downstream neuron fired ∼20% too fast in ours until this was matched. We found it by replaying the reference's own input spike times into our simulator and comparing every downstream spike.

Exact sparsity. Under strong visual drive ∼50,000 neurons sit just off rest, and stepping them made the brain four times slower than real time; parallelising the per-step pass across cores made it slower still (thread-region launch every 0.1 ms). Between inputs the model is linear, so the highest a neuron's voltage can still reach without further input has a closed form:

$$v_{\max} - v_0 \le \max(u, 0) + P_g \max(g, 0), \qquad P_g = \max_t \frac{\tau}{\tau - T}\left(e^{-t/\tau} - e^{-t/T}\right) \approx 0.157, \tag{1}$$

with $u = v - v_0$, T the membrane and τ the synaptic time constant. A non-refractory neuron whose bound is below threshold cannot spike before its next input; it becomes *dormant* and is not stepped. Its next synaptic arrival or Poisson event wakes it, and its state is advanced analytically over the gap (the per-step propagator raised to the gap, from a lookup table). Poisson input is drawn as geometric inter-event gaps per frame rather than a Bernoulli trial per neuron per step, which is the same process. The result remains spike-for-spike identical to Brian2 under replay (13,607 of 13,607 spikes) while cost follows synaptic events instead of neurons × steps.

2.2 Robot, simulator and interface

The microduck's production daemons (`robotd`, `tofd`, `mediad`) run unmodified against a MuJoCo [6] model of its body, with only the servo bus, the ToF sensor and the camera replaced by the simulator (`scripts/duck-sim`). Our bridge reads the robot's own interfaces: ToF frames at 15 Hz (`tof.stream`), camera frames at 30 Hz (`media.frame`), and the robot's state at 50 Hz (`robot.state`: head forward kinematics, IMU orientation, odometry, executed twist). It writes only intents the robot already accepts—`robot.move`, `robot.head`, `robot.look`, `robot.pose`, `robot.sound`. The brain advances by exactly the time between ToF frames, with the latest camera-derived rates.

A stimulus prop (a sphere that can be thrown at the robot) and a rotating drum of 48 alternating black and white panels (0.5 m radius, a 15° spatial period) are added to the simulator without modifying it: a generated copy of the scene, and a Python start-up hook that moves the two mocap bodies.

The twin had to walk first. The twin as shipped loaded an obsolete walking policy on a robot model without joint backlash, and did not walk: 4 cm in 8 s at 0.15 m/s commanded. The current policy (velstand) was trained with backlash and stands still without it; with both corrected the twin walks (0.06 m/s for 0.15 m/s asked) and turns (50 ř/s for 46 ř/s asked on the spot). This fix has been submitted upstream. The residual shortfall in forward speed, a dead band below ∼0.55 rad/s of yaw on the spot, and an occasional failure to take a first slow step are properties of the twin that our controllers have to live with.

Off the robot. On hardware the brain runs on a laptop and reaches the robot over the WebRTC session the robot publishes: the camera as an H.264 track and the robot's JSON-RPC interface over a data channel (`link.py`). WebRTC frames carry no capture time, so each is dated from the RTCP sender reports and a pair of robot clock readings onto the robot's monotonic clock, the one its state stream is stamped with; frame and state then pair without the network delay. Against the simulated robot's media daemon, following, the startle and the head reflex (gain 0.24 [0.19, 0.28], against 0.23 locally) work over this path.

2.3 Startle: depth sensor to looming detectors

The giant fibre (DNp01, one per side) receives its two largest excitatory inputs from LPLC2 and LC4 (10.6% and 8.6% of the left GF's annotated input synapses; 12.1% and 8.4% of the right's), consistent with their established roles in looming detection [9, 10, 8]. For each ToF zone we estimate inverse time-to-contact $\tau^{-1} = -\dot{d}/d$ from a line fit over four frames (0.27 s); a zone drives the LPLC2 and LC4 neurons assigned to it (left four columns to the left optic lobe, round-robin within a side, ∼3 LPLC2 and ∼1.5 LC4 per zone) at a saturating rate up to 150 Hz. Two filters are ours and labelled as such: zones whose return is the floor are excluded, using the robot's own floor test with a 3° pose margin; and drive is gated on approach speed above 0.4 m/s, a duck choice that loosely parallels the giant fibre's preference for fast looming [9]. A giant-fibre spike triggers a startle assembled from existing intents, with a 2 s refractory period.

2.4 Optomotor reflex: camera to motion detectors

Retinotopy from the connectome. Each eye has ∼750 T4 and ∼750 T5 neurons of each of four preferred directions (a front-to-back, b back-to-front, c up, d down [7]). FlyWire's column map [3] assigns 11,822 of them to 796 hexagonal columns (395 left, 401 right) with axial coordinates (p, q). Nearest neighbours differ by $(p{+}1, q{+}1)$ or by one coordinate, which gives physical coordinates $X = \sqrt{3}\,(q - p)/2$, $Y = (p + q)/2$ and a roughly round eye.

The map does not say which direction is anterior or dorsal. We derive both from the wiring (Fig. 3). LPLC2's four dendritic arms each take the T4/T5 subtype whose preferred direction points away from the cell's centre [8]; averaging, over all LPLC2 of an eye, the position of each subtype's inputs relative to the cell's input centroid places T4a/T5a at $\Delta X = -2.7$ and -2.9 columns (left and right eye), T4b/T5b at $+2.1$ and $+1.9$, T4c/T5c at $\Delta Y = +1.7$ and $+1.9$, and T4d/T5d at -2.1 and -2.2. Hence $+X$ is anterior and $+Y$ dorsal in both eyes. Independently, the dorsal and ventral HS cells (HSN, HSS) take their T4/T5 input

at higher and lower Y respectively, confirming the vertical axis, and the equatorial HSE sets elevation zero. With an interommatidial angle of 4.8° and the frontmost column placed 10° across the midline, the eyes span −10° to 131° in azimuth and −84° to 66° in elevation.

Flow to rates. The camera sees 45°×73° straight ahead. Only T4/T5 neurons whose direction falls in that view—1,424 of 11,822—receive input: dense Farnebäck optic flow [15] at quarter resolution, averaged over one column's extent at the neuron's position, projected on its preferred direction, rectified and mapped to a saturating Poisson rate (≤60 Hz). Image left is the robot's left (verified with a placed object).

Efference copies. Image motion the robot causes deliberately is predicted and subtracted before it reaches the fly: the head's idle look-around, and walking. Walking turned out to be dominated not by translation (floor flow ∼10 ř/s at 0.08 m/s) but by the gait's pitch and roll (30 ř/s to 80 ř/s). We predict both per pixel from the robot's state—floor translation from odometry on pixels whose ray meets the floor, rotation from the camera's world orientation between states with the vertical component removed (yaw slip is the reflex's own signal)—and subtract the prediction weighted by local image texture, since a flow estimator, like a T4 neuron, reports no motion on an edgeless surface. The prediction matches a controlled head nod with $r = 0.98$; during walking it correlates with the measured flow ($r = 0.90$) only when each frame is paired with the state at its own capture time.

Motor readout. DNp15 (DNHS1 [11]) left–right difference, smoothed (0.15 s), integrates into head yaw. For the body we tested DNa02 [12] directly (`-steer dna02`) and a head-led coupling of ours (`-steer body`): the walking body (0.1 m/s) turns at $k\cdot$ head yaw while the head counter-rotates by the trunk's measured yaw rate, so gaze is held.

2.5 Following: a camera target to the small-object pathway

FlyWire's brain is female, so it lacks P1 and the courtship-pursuit circuit. It does contain the visual object pathway that pursuit uses: LC10a, lobula columnar neurons tuned to small moving objects [18, 19], project to the anterior optic tubercle, whence AOTU019 (GABAergic, mostly onto the contralateral DNa02) and AOTU025 (cholinergic) reach DNa02. We locate each LC10a's receptive field as the synapse-weighted mean viewing direction of its inputs that lie on the column map (Tm20, Tm4 and others), on its own eye: 232 of 234 LC10a receive such input. The target—in simulation a small self-lit sphere, the only saturated object in the grey camera image, found as the largest bright blob after a morphological opening—gives a bearing and elevation through the pinhole model, and drives each LC10a at $r_{\max}\exp(-d^2/2\sigma^2)$, d the angle between target and receptive field ($r_{\max} = 120$ Hz, $\sigma = 10°$, chosen in a dose sweep in which DNa02 is graded; at 250 Hz it saturates and folds). The readout is DNa02's left–right difference, smoothed (0.4 s), as yaw rate while walking at 0.1 m/s; with no target in view the robot stands, and the head is held straight so that the camera faces where the body goes. In closed loop, a target on the floor 0.45 m ahead of the robot swings round it at 8 ř/s, 90° left, back, 90° right and back (65 s), scored from the simulator's ground truth.

2.6 Following a person: a motion encoder for a real room

The self-lit target is a stand-in: on video from a phone at the robot's height (four clips, a walk round the phone, an empty room, the phone turning in it and onto the walker's feet; not published) no pixel reaches the detector's threshold, and a lower threshold locks onto a glint above the equator. For a real room the encoder looks for motion, not brightness, and picks no target. *Motion*: each frame against the one 0.1 s earlier, the camera's own turn between their capture times (from the robot's state history, as the frames are dated) and its vertical bob (phase correlation) taken off; flicker, thin edges, blobs below 0.2% or above 35% of the frame, scenery that slides the way of the camera's turn but further (parallax), and, while the camera turns faster than 20 ř/s, everything (as flies damp vision during saccades) removed. Of each moving blob only its lowest 8° drives: where a walking person meets the floor. *Pooling*: the motion image is pooled into 2° cells and each LC10a is driven by the moving area under its receptive field, Gaussian-weighted as above; a moving disc of 2.5° at its centre drives it at $r_{\max}$, and the drive decays with 0.3 s when the motion stops. *Readout*: the yaw rate is the left–right difference of the four descending-neuron pairs that followed a target's bearing most steeply open loop (Results; DNa02, DNae002, DNa16, and DNg97 with its sign inverted), less a lean in proportion to the total LC10a drive, calibrated for each brain on synthetic targets dead ahead (0.0036 intact, 0 rewired), smoothed (0.4 s) and scaled; the robot turns on the spot; a turn away from the world heading where the target was last seen, beyond 10°, is suppressed (a heading memory); and a flinch whose looming peaks within 20° of the target being followed is withheld (the giant fibre still fires and is counted). The test stimulus is a pair of mocap legs, dark trousers and grey feet, stepping 0.6 m from the robot along the

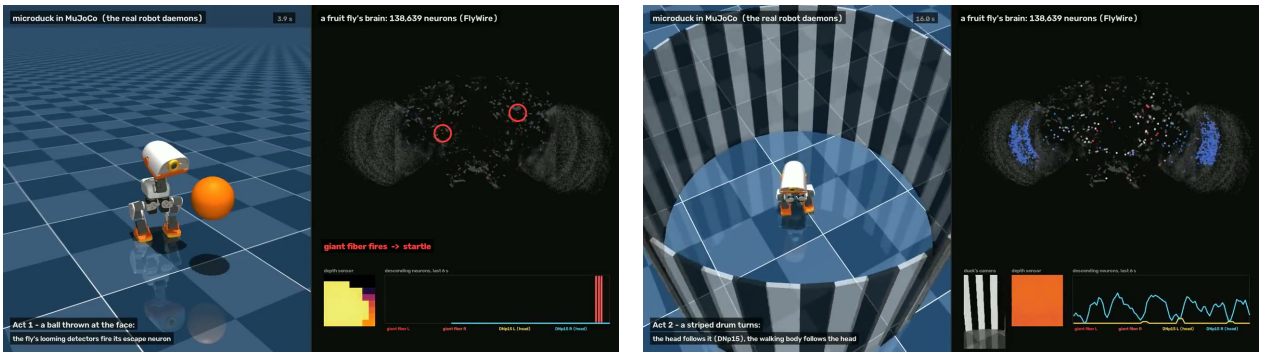


Figure 1: Frames from the recorded demonstration (`media/duckophila.mp4`). Left halves: the robot in MuJoCo, re-rendered from its recorded pose. Right halves: the brain, every neuron at its FlyWire position seen from the front, lit in its class's colour when it spikes. **Left:** a ball reaches the robot's face; both giant fibres are ringed in red as they fire. **Right:** in the turning drum, T4/T5 neurons in both optic lobes (blue) and DNp15 respond.

same 90° schedule, on a non-repeating plank floor generated for the purpose (the simulator's checkerboard is periodic, which defeats image-shift estimates, and its bright lines shimmer into motion as the robot rocks).

2.7 Learning: what we had to add

Cues. The model's mushroom body does not give a cue its own Kenyon cells (Results), so each cue—here the ball or an upright bar, which the detector tells apart by the blob's aspect ratio—is given a random 10% of the 5,177 Kenyon cells, driven at 20 Hz, as models of the mushroom body assume random projection-neuron convergence [20].

Plasticity. The connectome is static. After each brain step, every Kenyon cell → MBON weight is multiplied by $1 - \eta\, a_{\mathrm{KC}}\, d_{\mathrm{MBON}}$: a whether the Kenyon cell fired, d the rate of the dopamine neurons of the MBON's compartment relative to 10 Hz (clipped at 1), $\eta = 0.5$ per second of pairing—dopamine-gated depression [21, 22]. The compartment is the connectome's: the MBON's direct DAN inputs, weighted by synapse count (every MBON has some; the pairings agree with the literature, e.g. PAM11 with MBON07).

Valence and reward. Petting drives all 307 PAM neurons at 30 Hz (reward); PPL1 would be punishment. An MBON's valence is the opposite of its compartment's dopamine [23, 24]: mostly PAM, avoidance; otherwise approach (40 and 56 MBONs). A cue's value is the mean approach-MBON rate minus the mean avoidance-MBON rate.

Lesion. The antennal lobe's 1,152 local, projection, input and output neurons are silenced (outgoing synapses set to zero) for learning, since the runaway starts there and the robot has no olfactory input; it leaves following (DNa02 by bearing) and the startle (10 of 10 at 1 m/s) unchanged.

Behaviour. In closed loop the robot follows a target (as above) only while its smoothed value exceeds its naive value—the mean over its first 8 s in view—by 1 Hz; otherwise it stands. The robot detects petting acoustically (an on-board classifier of head-scratch sounds) but does not yet publish it, so in simulation a script sends it.

3 Results

3.1 The stepped model reproduces the reference

With the sugar-sensing gustatory neurons driven at 150 Hz for 30 one-second trials, firing rates of the 450 active neurons correlate with the Brian2 reference at $r = 0.9998$; MN9 fires at 83.2 Hz against 83.9 Hz (Fig. 2). Under replay of the reference's own input spikes, all 13,607 downstream spikes are identical in neuron and time step. On one core of an Apple M5 Max, one second of brain takes 0.04 s under that stimulus, 0.38 s under drum-like drive of 3,000 T4/T5 at 30 Hz, and 0.78 s with all 12,246 T4/T5 at 30 Hz; in the closed loop, a 67 ms frame costs 4 ms to 7 ms.

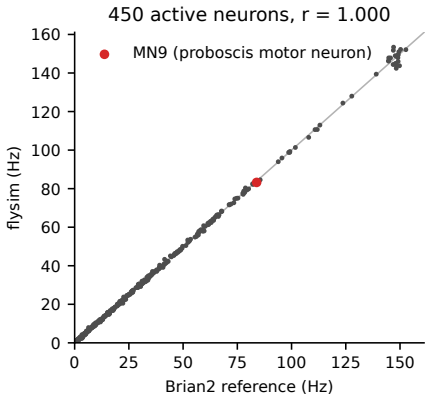


Figure 2: The stepped simulator against the Brian2 reference: mean firing rates over 30 trials with the right sugar-sensing neurons at 150 Hz (FlyWire 630, as in [4]).

3.2 Startle

Driving LPLC2 and LC4 of one side directly, the giant fibres are silent at rest and fire in a graded way with drive rate and with the fraction of the population driven, more strongly ipsilaterally; at higher drive the contralateral DNa01/DNa02 and, weakly, the moonwalker descending neurons (MDN [13]) are recruited. On synthetic ToF scenes ray-cast with the sensor's beam geometry and noise (a 5 cm ball approaching from 1.2 m, left or right, 40 trials per speed), the giant fibre fired in 0/40 trials at 0.2 m/s, 21/40 at 0.4 m/s, 40/40 at each of 0.6 m/s to 1.5 m/s, and 28/40 at 2.0 m/s, with a mean latency of 0.54 s from onset for approaches $\geq$1 m/s. Through the full robot stack in the simulator, 20 of 20 balls at 1.5 m/s produced a startle (1.00, 95% CI [0.84, 1.00]), latency 0.43 s, and 0 of 20 hands at 0.2 m/s (Table 2). The speed tuning is largely ours: the approach-speed gate imposes the cut, and the input null below shows the same cut. Very fast approaches are missed 30% of the time by the encoder, and walking towards a wall at 0.3 m/s produced a false startle in 6 of 20 trials.

Two failure modes were found and fixed in the interface, not the brain. A standing robot's idle head motion makes $\sim$45 of 64 depth zones see an approaching and receding floor; without a floor filter 30% of quiet frames drove the looming detectors. Near the horizon, a 2° to 3° discrepancy between the kinematic model and the rendered sensor pose decides whether a grazing return is floor; without a pose margin, the robot startled 5 times in 25 s in an empty room, with it none.

3.3 Optomotor reflex

In the connectome. Driving T4a/T5a of one eye and T4b/T5b of the other—the pattern a yaw rotation of the world produces—activates the ipsilateral HS cells and contralateral H2 as in the fly, and flips the left–right activity of DNa02 with the direction of rotation, in the sense of a syndirectional (following) turn under the interpretation of [12]: world turning right, DNa02 right 28 Hz versus left 5 Hz; world turning left, left 69 Hz versus right 5 Hz. DNa01 does not flip cleanly. The strongest-flipping descending neuron is DNp15 (DNHS1).

Retinotopy is necessary. With T4/T5 neurons assigned to image regions at random, wide-field drum motion reached LPLC2 scrambled and fired the giant fibre 7–11 times per run. With column-based placement, rotation produced no giant-fibre activity in synthetic tests or in the closed loop (one startle in 29 s of drum, from the ToF at start-up).

The camera's field of view limits the readout. With the column map, a whole-eye yaw rotation flips both DNa02 and DNp15 (Fig. 4). Restricted to the neurons whose direction falls in the camera's view, DNp15 still flips strongly (51 Hz left vs. 0 Hz right; 0 vs. 86 Hz), but DNa02 does not change sign: right exceeds left in both directions, by 2.7 Hz and 11 Hz. Rotation over the frontal 30° of the eye at all elevations is sufficient for a DNa02 flip; the camera's limited elevation range is not. Likewise, no expansion stimulus within the camera view, up to 400 ř/s, drove LPLC2 appreciably; looming therefore remains the ToF's task.

Closed loop. With DNp15 driving head yaw, the robot's head follows the drum in both directions as a triangle wave at about 7 ř/s for a 30 ř/s drum (closed-loop gain 0.23, 95% CI [0.21, 0.27], over 12 drum

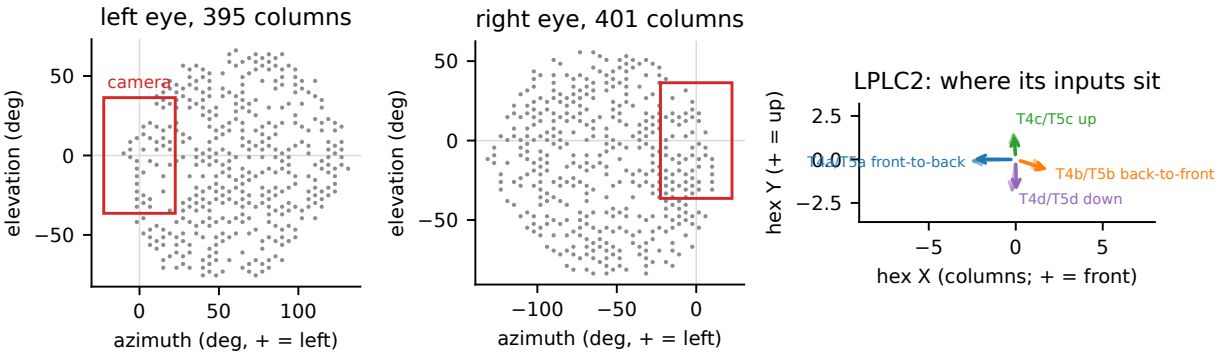


Figure 3: Retinotopy from the connectome. **Left, centre:** viewing directions of the 395 and 401 columns of each eye, from FlyWire's column map with orientation derived from the wiring; red, the camera's field of view. **Right:** mean offset of each T4/T5 subtype's input to LPLC2 from the LPLC2 neuron's input centroid (solid: left eye; faint: right). Front-to-back inputs sit posterior, up inputs dorsal—the outward-motion arrangement of a looming detector—which fixes the eye's axes.

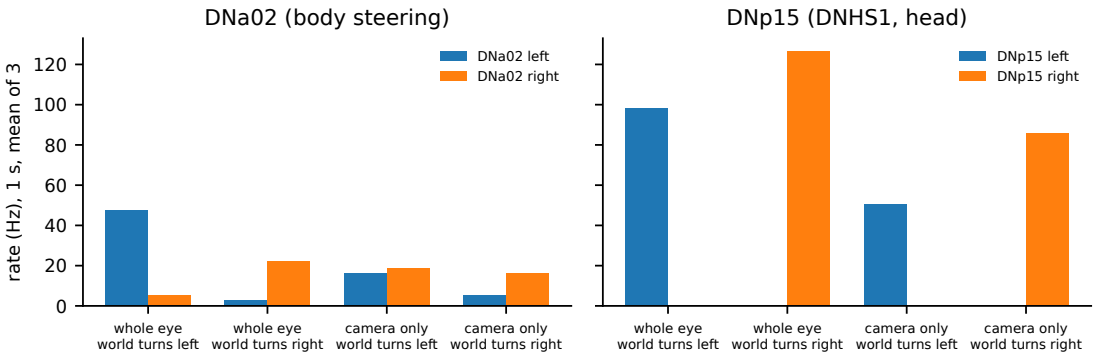


Figure 4: Steering descending neurons under yaw rotation (40 ř/s), with every T4/T5 driven (whole eye) or only those looking into the camera's view. DNp15 flips in both cases; DNa02 only with the whole eye.

phases: partial compensation) and holds still when the drum is still; higher gains oscillate, given ∼0.3 s of loop latency. With the head-led body coupling, the walking robot turns with the drum at a mean of +5 and −9 ř/s (peaks ±15 ř/s to 20 ř/s) against an intrinsic gait drift of −4.6 ř/s. Without the walking efference copy, three startles occurred in 200 s of walking in the turning drum; with it, none, with unchanged steering.

3.4 Following

Open loop. A target 10° below the eye's equator makes DNa02 turn toward it, graded with bearing: left minus right −5.5, −1.8, +1.6, +5.3 and 14.2 Hz at −20, −10, 0, 10 and 20° (+ = left; 20 seeds; Fig. 5). Across 383 paired descending neuron types the left–right difference of nine tracks the bearing ($r > 0.5$); DNa02 ranks third ($r = 0.94$), after DNae002 and DNa16. The rewired brain gives no DNa02 response, and with the same rates sent to as many random other visual projection neurons the bearing dependence vanishes ($r = 0.05$). Two properties of the pathway shape the behaviour. It responds to targets below the equator only: the turn toward a target at ±10° is 0 Hz at 5° elevation and above, 2.1 Hz at −5° and 3.9 Hz to 4.4 Hz from −10° down; a target at the camera's height, which appears above the image centre, is ignored, so the target is on the floor. And it is asymmetric: the left eye's LC10a receptive fields lie further anterior than the right's (a target dead ahead drives the left population at 931 Hz in total against 782 Hz), so DNa02 leans left and a target to the right gets about a third of the turn of one to the left. Whether this asymmetry is the fly's or our column-to-angle approximation we cannot tell.

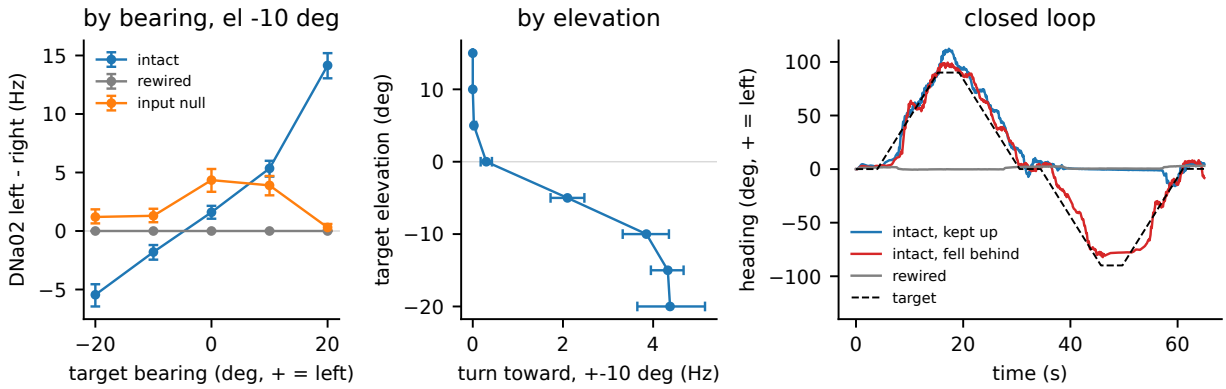


Figure 5: Following. **Left:** DNa02 left minus right against the target's bearing, open loop (mean, 95% CI, 20 seeds), for the intact brain, the rewired brain, and the input null. **Centre:** the turn toward a target 10° out, against its elevation. **Right:** closed loop, the robot's heading against the target's for a run that kept up, one that fell behind after a reversal, and the rewired brain.

Closed loop. In 4 of 8 runs the robot kept the target in view throughout, with a heading gain of 1.00 and a median error of 7.8° [7.4, 8.2]. In two of the other four it trailed just outside the camera's half-width (at most 29°) from the first reversal on and kept turning with the target (gain 1.00 and 0.94); in two it fell further behind after a reversal and lost the target for part of the run (gain 0.56 and 0.69). Over all eight runs the mean heading gain was 0.90 [0.78, 1.00]. The rewired brain never followed (0 of 8, 95% CI [0.00, 0.32]; gain −0.01). No run of either brain produced a startle.

A person's feet. Open loop, synthetic motion at 15° to the left through the pooling (with a 5° reference disc) turned DNa02 by 10.7 Hz for a ball-sized patch and 12.3 Hz for two feet low in the view, but by 0.3 Hz for the same feet with whole legs above them to 30°: too much of the small-object pathway driven at once no longer steers, which is why only each blob's lowest part drives it. On the stepping legs in closed loop, DNa02 alone did not steer: it fires a few spikes a second, and its left–right difference (4.8 Hz with the person on the left, 2.3 Hz ahead, 0.1 Hz on the right) drowned in their noise. Of 383 paired DN types, DNae002 and DNa16 follow a target's bearing more steeply than DNa02 (9.5 and 6.4 against 4.6 Hz per 10°; DNg97 −3.5), and their combined difference changed sign between the sides (36 Hz left, −12 Hz right) but leant left, 13.5 Hz with the person dead ahead—in proportion to the LC10a drive, and absent in the rewired brain, whose readout neurons do not respond. Each further rule answered a measured failure: walking toward the person made the floor flow through the lower field and the intact brain followed its own flow round in circles; a turn the DNs kept up after a fast turn of the robot's own carried it into a spin; and the legs, 0.6 m away, loomed on the depth sensor and fired the giant fibre ∼8 times a run, each flinch breaking the following. With all of them (Methods), over 8 runs the intact brain kept a median heading error of 19° [13, 28] and the person in view 59% [47, 69] of the time (heading gain 0.62 [0.41, 0.79]; 7 of 8 runs followed, the best at 8.4° and 80%); the rewired brain 40°, 35% and 0.02—the share a robot that does not turn keeps in view. The glowing ball, for comparison: 10°, 84%, 0.90. On the four phone clips, open loop, the encoder and brain turned toward the walking person 87% of the time they turned, and not at all in the empty room, standing or turning. Integrate-and-fire saccades on the same readout [25] also followed but overshot; a single equator for both eyes (HSE's input centroid, one cell per eye, puts them ∼10° apart on an otherwise identical column grid) did not remove the lean.

3.5 The mushroom body, and learning

No code of its own. Kenyon cells take visual input mainly in KCg-d (30% of their input synapses, from aMe12 and medulla/lobula projection neurons; KCab-p 3%, others <1%). Nothing the camera can drive reaches them: rotation on the camera's T4/T5, the follow target on LC10a, a 10° patch on every columnar neuron, and all 45,528 columnar neurons at 30 Hz each fire none of the 31 visual projection neurons that feed Kenyon cells, which take only 3% of their input from column-mapped neurons. More fundamentally, the model's Kenyon cells do not code their input. Four odours (one olfactory receptor neuron type each) and taste either reach no Kenyon cell or the same ∼3,370 of 5,177 (Jaccard overlap 0.99–1.00), with the whole brain at ∼470 000 spikes/s; a fly uses a sparse, odour-specific 5–10%. Removing the Kenyon cells' synapses onto each other (38% of their output weight) leaves it so (overlap ≥ 0.98). The runaway begins in the antennal lobe,

Table 2: Controls. Open loop: 20 seeds per condition (40 trials per speed across both sides); closed loop: 20 throws, 8 following and 6 learning runs, through the full robot stack. 95% Wilson (proportions) or bootstrap (means) intervals.

	intact	rewired	sign-flipped	input null
Startle, 0.6 m/s to 1.5 m/s, open loop	1.00 [0.91, 1.00]	0.00	0.00	0.50 [0.35, 0.65]
Startle, 0.2 m/s, open loop	0.00 [0.00, 0.09]	0.00	0.00	0.00
Startle, 1.5 m/s, closed loop	1.00 [0.84, 1.00]	0.00 [0.00, 0.16]		
Startle, 0.2 m/s, closed loop	0.00 [0.00, 0.16]	0.00 [0.00, 0.16]		
Ipsilateral GF spike share	0.58 [0.56, 0.59]	–	–	0.54 [0.52, 0.56]
False startle, walking at a wall 0.3 m/s	6/20	0/20	0/20	2/20
Head optomotor gain, closed loop	0.23 [0.21, 0.27]	0.00		
DNp15 follows rotation (camera view)	yes, rank 1/383	no	no	
DNa02 follows rotation	whole eye only	no	no	
DNa02 tracks target bearing, open loop	$r = 0.94$, rank 3/383	$r = 0.00$		$r = 0.05$
Following, runs kept up, closed loop	4/8 [0.22, 0.78]	0/8 [0.00, 0.32]		
Following feet, median error, closed loop	19° [13, 28]	40°		
Learning index, open loop (Hz)	+3.35 [2.83, 3.82]	0.00		
Walks after petted / other target	5/6 / 0/6	0/6 / 0/6		

whose local and projection neurons fire just before the Kenyon cells do; with the antennal lobe silenced, a random 15% of Kenyon cells at 20 Hz never ran away (0 of 10, intact 3 of 10). In this model the dopamine neurons, like every other synapse whose transmitter is not GABA or glutamate, are excitatory.

Open loop. With the additions above, pairing cue A four times with petting raised its value over that of an unpaired cue B by 3.35 Hz [2.83, 3.82] (10 seeds). The same cue and the same petting on separate trials (−0.39 [−0.94, 0.14]) and no petting (−0.12 [−0.70, 0.45]) did not; in the rewired brain nothing changed (0.00). With valence read off the MBONs' transmitters instead (glutamate avoidance, acetylcholine and GABA approach [23]), petting made the paired cue *less* attractive, almost entirely through the GABAergic MBON09 in a PAM compartment.

Closed loop. Six runs per brain, the petted target alternating between ball and bar. Naive, the robot followed neither (0 of 6 each). After four 3-s pettings with one target in view and four presentations of the other, the petted target's value had risen by 3.8–4.2 Hz and the other's by 0.3–0.6 Hz in all six runs; the robot then walked after the petted target in 5 of 6 runs (net 1.03 m [0.70, 1.26] in 30 s; heading gain 0.76 [0.41, 0.96]; ball and bar both) and after the other in none. The failure learned but lost the target while following. The rewired brain learned and followed nothing.

3.6 Controls

To ask what the connectome contributes, we repeated each experiment with brains that are wrong in one respect (Table 2): a *rewired* connectome, in which every connection keeps its presynaptic neuron and weight but receives a target from a random permutation of all targets (so every neuron keeps its in- and out-degree); a *sign-flipped* connectome; an *input null*, the intact brain with the looming encoder's output sent to as many randomly chosen other visual projection neurons of the same side; and a *readout null*, every descending neuron type with exactly one neuron per side scored for following rotation.

The rewired and sign-flipped brains remain active—55 000 spikes per second under visual drive, against 66 000 intact—but activity no longer reaches the descending neurons: no startle in any of 280 open-loop or 20 closed-loop trials, no DNp15 or DNa02 response, and a closed-loop head gain of 0.00. DNp15's following is specific: of 383 paired descending neuron types, 13 follow a rotation within the camera's view (95% CIs excluding zero in both directions) and DNp15 follows most strongly (rank 1, as also over the whole eye). The escape response is less specific than the demonstrations suggest: the input null still fired the giant fibre in about half of fast approaches (0.50 [0.35, 0.65] at 0.6 m/s to 1.5 m/s, against 1.00 for LPLC2/LC4), so in this model the giant fibre can be reached from visual input in general, and LPLC2/LC4 make it reliable. Laterality is weak: the giant fibre ipsilateral to the ball carried 58% of giant-fibre spikes (95% CI [56, 59]) in open loop and 57% [45, 66] in closed loop, close to the input null's 54% [52, 56].

4 Discussion

The connectome, unmodified and untrained, produced three recognisable fly behaviours when its sensory neurons were driven by a robot's sensors: an escape decision to fast approaching objects, a following optomotor response, and turning toward a small object. None was wired in, and all disappear when the same neurons and weights are rewired. Their specificity differs: the optomotor readout singles out DNp15 among hundreds of candidates, whereas the escape decision is made more reliable, but not made possible, by the looming detectors, and carries little information about the side of the threat. An earlier, demonstration-based version of these results described the escape as lateralised; the controls do not support that. What was wired in is listed in Table 1, and most of the effort went there: the biological plausibility of the input decided whether the connectome's computation could be expressed at all. Scrambled retinotopy did not just add noise, it produced false escape decisions, because LPLC2's selectivity is a property of where its inputs sit. The robot's own movement dominated its visual input, as it does for animals, and needed an efference copy before the fly's motion pathway could be used while walking [14].

The clearest limitation is sensory. A single forward camera covers a small part of a fly's visual field, and the canonical steering neuron DNa02 [12] did not respond to rotation within it. The head-coupled neuron DNp15 did, so we steer the head with the fly's output and the body with a coupling rule of ours. Whether a wider camera, or several, recover DNa02-based steering is a direct experimental question the setup can answer. For a small target DNa02 does respond within the camera's view, so following is steered by the fly's own steering neuron; but its losses—a target that leaves the 45° view is not found again—are again the camera's, against a fly's nearly panoramic eyes.

Other limitations are those of the model and the platform. The LIF model of [4] uses one synaptic weight for all connections, has no gap junctions (the giant fibre's electrical synapses are absent), no plasticity and no neuromodulation, and treats neurotransmitter predictions as signs. T4 and T5 are driven alike, without an ON/OFF split. The robot is simulated; its gait reaches 40% of commanded speed and the camera/state timing measured in simulation will differ on hardware. Our gains and gates are engineering choices, stated as such. With the self-lit target, LC10a's preference for moving over stationary objects is not modelled; with the motion encoder only what moves drives it. We make no claim that the robot behaves like a fly beyond the three behaviours measured.

Following a person shows where the boundary moves when the input is real. The pathway is the fly's—two feet low in the view steer DNa02 as a target does, the whole legs do not, and nothing follows when the wiring is rewired—but steering from it took more of ours than following a glowing target did: a readout over four DN pairs instead of one, a correction for a lean that appears in our receptive-field map, and rules for a body that moves its own camera (turning on the spot, a heading memory, suppression during its own fast turns, no flinch at what it pursues). Several have counterparts in flies, population readout of descending neurons, saccadic suppression, a compass in the central complex, but here they are rules, not circuits. The lean is likely ours as well: the two eyes' maps differ by $\sim 10^\circ$ in elevation where their column grids do not, which a better placement of each eye's equator than one cell's input centroid should settle.

Learning is different in kind. The model's mushroom body does not code its input, so the cue codes, the plasticity, the valence assignment and a lesion are ours, and what the connectome contributes is narrower: the Kenyon cell $\rightarrow$ MBON wiring and its compartments, which carry a cue-specific, pairing-specific change through to behaviour and fail to when rewired. Why the reference model's antennal lobe runs away—one shared synaptic weight, no graded inhibition, excitatory neuromodulators—is a question for the model, and a prerequisite for using it to study learning.

The natural next step is the physical robot. The path to it is in place: the brain on a laptop, reaching the robot over the WebRTC session it publishes, with frames dated on the robot's clock (Methods); what remains is to measure the real camera's delay, as was done in simulation. Related efforts embody connectome-constrained or neuromechanical fly models in simulated bodies [16, 17]; here the body is a different animal's, and the interesting part is the translation between them.

Reproducibility

Everything above is a script in `https://github.com/first-it-consulting/duckophila`: `experiments/` (each figure and number), `paper/figures.py` (every figure, from data), `demo.sh -video` (the recorded demonstration). The FlyWire data are fetched under their own terms. Hardware: Apple M5 Max, macOS; brain on one core.

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